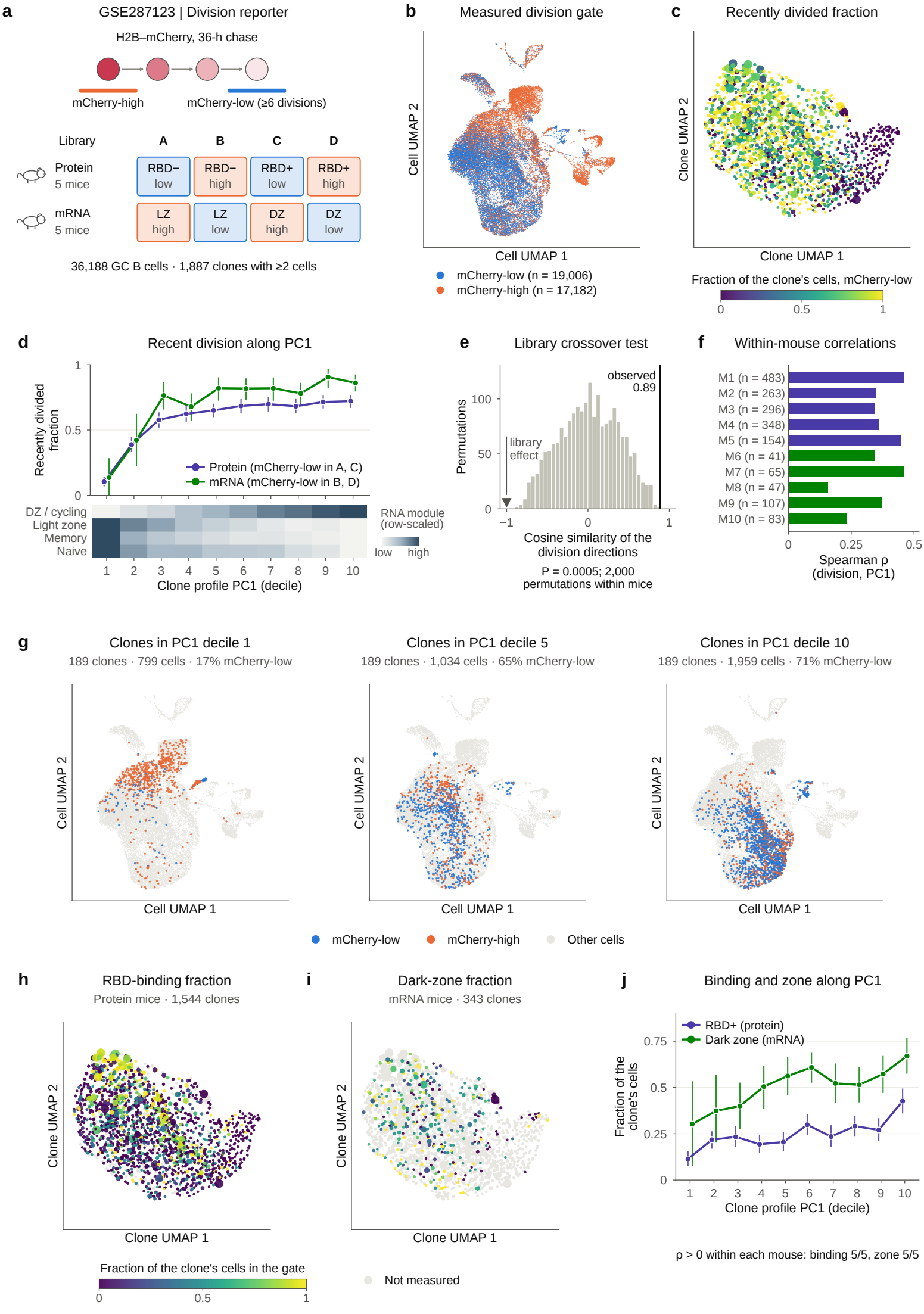

a
GSE287123 | Division reporter
H2B–mCherry, 36-h chase
mCherry-high
mCherry-low (≥6 divisions)
Library
A
B
C
D
Protein
5 mice
RBD− low
RBD− high
RBD+ low
RBD+ high
mRNA
5 mice
LZ high
LZ low
DZ high
DZ low
36,188 GC B cells · 1,887 clones with ≥2 cells
b
Measured division gate
Cell UMAP 2
Cell UMAP 1
mCherry-low (n = 19,006)
mCherry-high (n = 17,182)
c
Recently divided fraction
Clone UMAP 2
Clone UMAP 1
Fraction of the clone's cells, mCherry-low
0
0.5
1
d
Recent division along PC1
Recently divided fraction
Protein (mCherry-low in A, C)
mRNA (mCherry-low in B, D)
DZ / cycling
Light zone
Memory
Naive
RNA module (row-scaled)
low
high
Clone profile PC1 (decile)
e
Library crossover test
Permutations
observed 0.89
library effect
Cosine similarity of the division directions
P = 0.0005; 2,000 permutations within mice
f
Within-mouse correlations
M1 (n = 483)
M2 (n = 263)
M3 (n = 296)
M4 (n = 348)
M5 (n = 154)
M6 (n = 41)
M7 (n = 65)
M8 (n = 47)
M9 (n = 107)
M10 (n = 83)
Spearman ρ (division, PC1)
g
Clones in PC1 decile 1
189 clones · 799 cells · 17% mCherry-low
Clones in PC1 decile 5
189 clones · 1,034 cells · 65% mCherry-low
Clones in PC1 decile 10
189 clones · 1,959 cells · 71% mCherry-low
Cell UMAP 2
Cell UMAP 1
mCherry-low
mCherry-high
Other cells
h
RBD-binding fraction
Protein mice · 1,544 clones
Clone UMAP 2
Clone UMAP 1
Fraction of the clone's cells in the gate
i
Dark-zone fraction
mRNA mice · 343 clones
Not measured
j
Binding and zone along PC1
RBD+ (protein)
Dark zone (mRNA)
Fraction of the clone's cells
Clone profile PC1 (decile)
ρ > 0 within each mouse: binding 5/5, zone 5/5